

The Role of Cosmetic Attributes in Used-Car Price Formation: Evidence from the Turkish Second-Hand Market

Final Report

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GitHub Repository: [derobtw/derinalc_dsa210](https://github.com/derobtw/derinalc_dsa210)

Abstract

This report investigates whether cosmetic attributes, especially vehicle color, materially affect used-car prices in the Turkish second-hand market after structural vehicle characteristics are taken into account. The analysis uses an arabam.com-based Kaggle dataset containing 50,755 listings. After cleaning, filtering, and outlier trimming, 48,062 observations remained for analysis. To enrich the dataset, official 2026 Motor Vehicle Tax (MTV) tariff information from the Revenue Administration of Türkiye (GİB) was converted into an estimated annual ownership-cost variable, allowing tax estimates to be assigned to 46,947 vehicles. Exploratory data analysis showed that prices are strongly right-skewed, that mileage is negatively associated with price, and that structural attributes such as transmission and body type visibly segment the market. Statistical testing found that raw price distributions differ across color groups (Kruskal–Wallis statistic = 4810.14, $p < 0.001$). A nested OLS comparison also showed that color is statistically significant after controlling for structural variables ($p = 2.14 \times 10^{-14}$), but the gain in adjusted R^2 was only 0.00014. Machine learning comparisons reinforced this conclusion: adding color produced only a marginal improvement in linear regression and worsened random forest performance. Overall, the findings suggest that color is statistically detectable but practically weak; the main drivers of used-car prices are structural features such as brand, series, model year, mileage, transmission, fuel type, and body type.

1. Motivation and Research Question

Used-car markets are rich in consumer intuition. Buyers and sellers often believe that visible cosmetic attributes — especially color — can substantially influence resale value. At the same time, price formation is also shaped by more fundamental vehicle characteristics such as brand, model family, production year, mileage, fuel type, transmission, and body type. This project was designed to test whether cosmetic variables meaningfully contribute to price once those structural determinants are controlled for.

The main research question is: To what extent do cosmetic attributes influence used-car prices after controlling for structural vehicle characteristics in the Turkish second-hand car market?

The working hypothesis is that cosmetic attributes, particularly color, have only a limited marginal effect on used-car prices once the main structural variables are taken into account.

2. Data Sources and Collection

The primary dataset is a publicly available used-car listing dataset hosted on Kaggle and based on arabam.com listings. It includes information such as price, brand, series, model, production year, mileage, transmission type, fuel type, body type, color, engine volume, engine power, and seller type. The raw dataset contains 50,755 observations.

The course guideline required enrichment when using a public dataset. The original proposal considered time-linked macroeconomic variables such as exchange rates or fuel prices. However, the listing dataset does not include listing dates, which makes observation-level time matching weak. For this reason, the

external enrichment source was revised to official 2026 Motor Vehicle Tax (MTV) brackets published by the Revenue Administration of Türkiye (GİB).

Vehicle age was derived from production year, engine-volume brackets were created using engine displacement, and official MTV tariff categories were used to assign an estimated annual MTV value. Because the exact legal vehicle-value field used in the GİB tariff is not available in the listing dataset, listing price was used as a proxy for the value bracket. As a result, the constructed variable should be interpreted as estimated annual MTV rather than the exact legal tax liability.

Dataset item	Value
Raw observations	50,755
Cleaned observations used for EDA and testing	48,062
Listings assigned estimated MTV	46,947
Main market source	arabam.com-based Kaggle dataset
External enrichment source	Official 2026 GİB MTV tariff

Table 1. Summary of the data sources and cleaned analytical sample.

3. Data Preparation and Feature Engineering

Column names were standardized into analysis-friendly English labels. Numeric variables such as price, mileage, year, engine volume, and engine power were converted into numeric format where needed. Duplicate records were removed, and observations with missing values in the main analytical variables were excluded. Rows with invalid prices, negative mileage values, or unrealistic years were filtered out.

To reduce the impact of extreme observations, the price and mileage variables were trimmed using the 1st and 99th percentiles. Because both variables were right-skewed, logarithmic transformations were created: `log_price` and `log_km`. For later modeling, rare categories were grouped into “Other” for selected categorical variables such as color, body type, and series. This made both the statistical models and the machine learning pipelines more stable.

The estimated MTV enrichment introduced three derived concepts: vehicle age, engine bracket, and a proxy value bracket. The final estimated MTV variable is meaningful as an ownership-cost proxy, but it was not used as a predictor in the machine learning price models because its construction partially depends on listing price and would introduce leakage.

4. Exploratory Data Analysis

Exploratory data analysis was used to understand distributions, inspect relationships, and assess whether visible raw patterns were consistent with the project hypothesis. Four patterns were especially important.

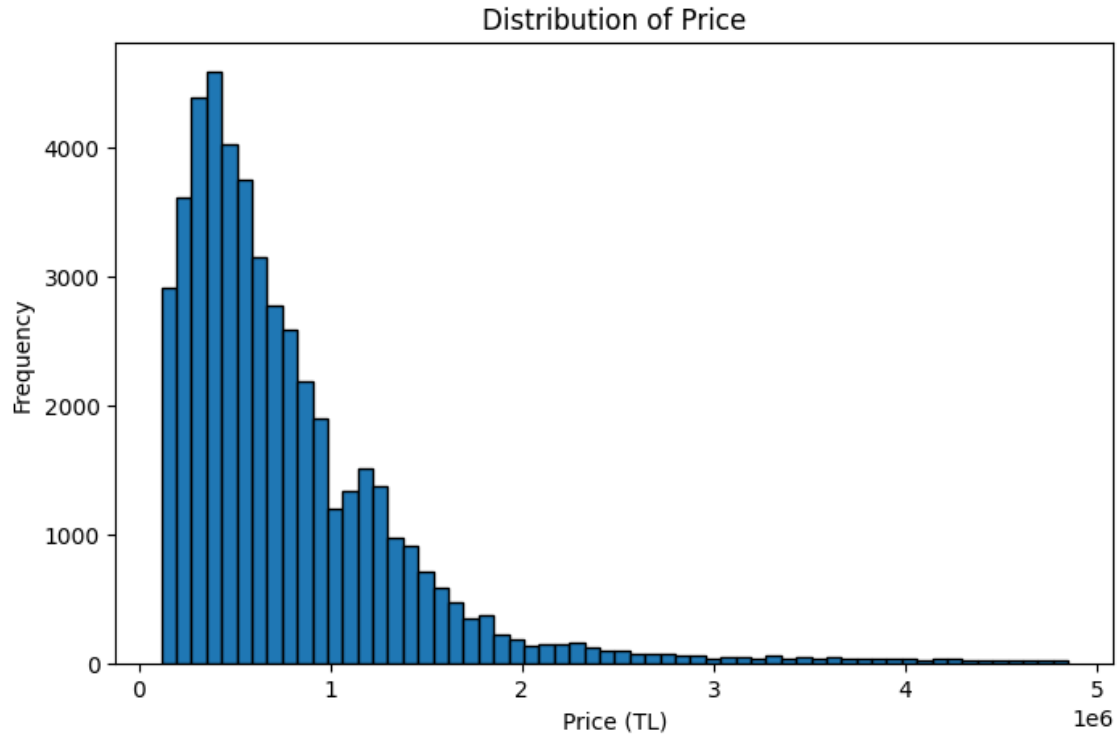


Figure 1. Distribution of used-car prices. Prices are strongly right-skewed, which justifies using log-transformed price in later analysis.

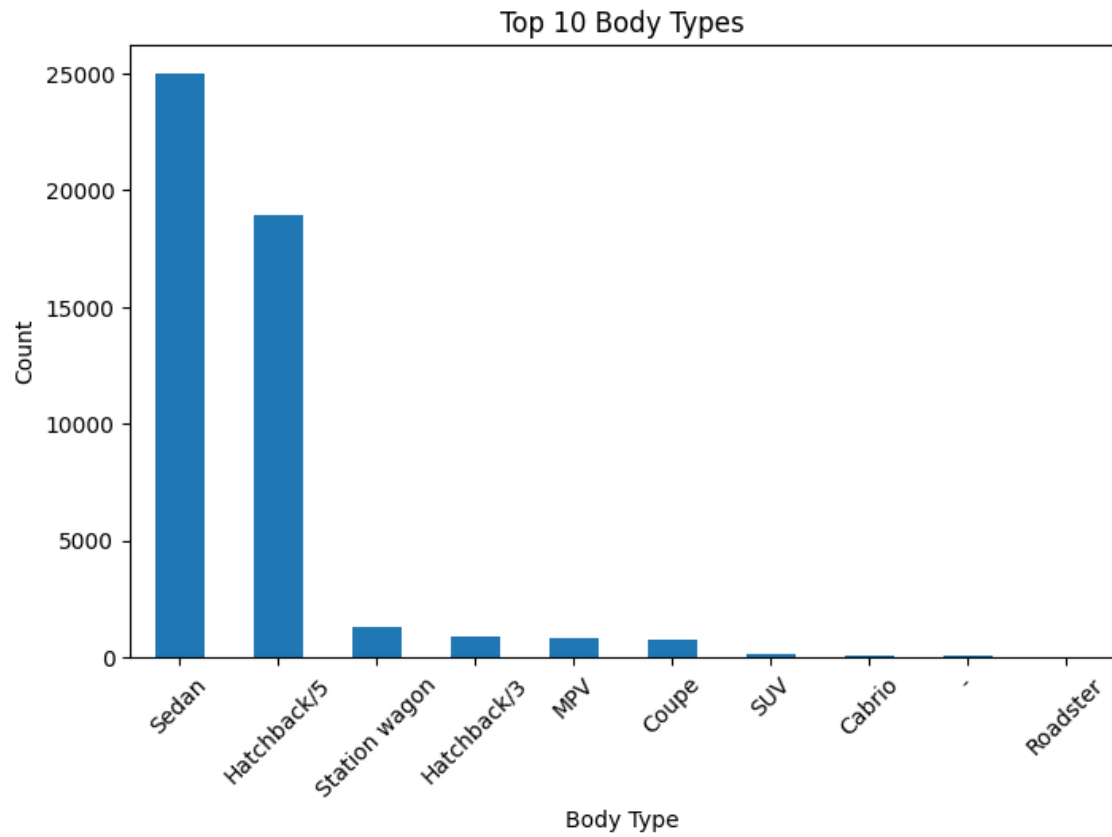


Figure 2. Price versus mileage. The relationship is generally negative, although there is still substantial dispersion within mileage bands.

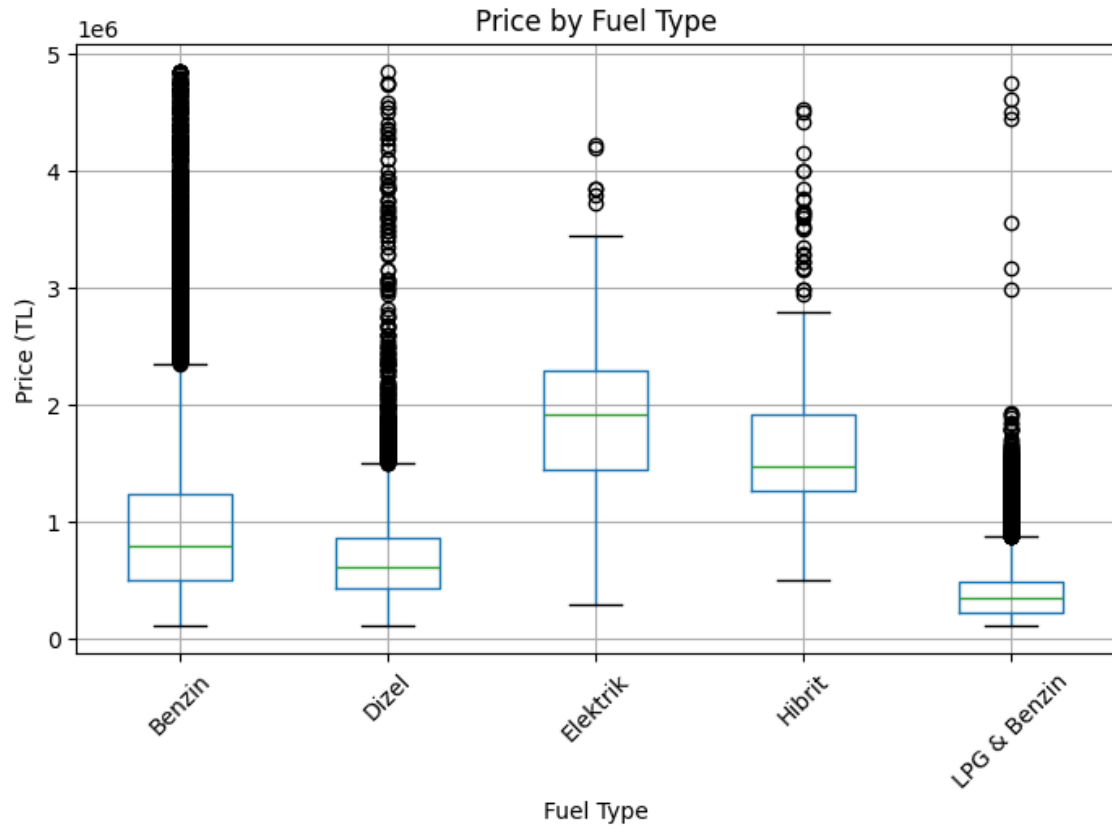


Figure 3. Price by color group. Raw differences across colors are visible, but they may reflect structural differences rather than a pure color effect.

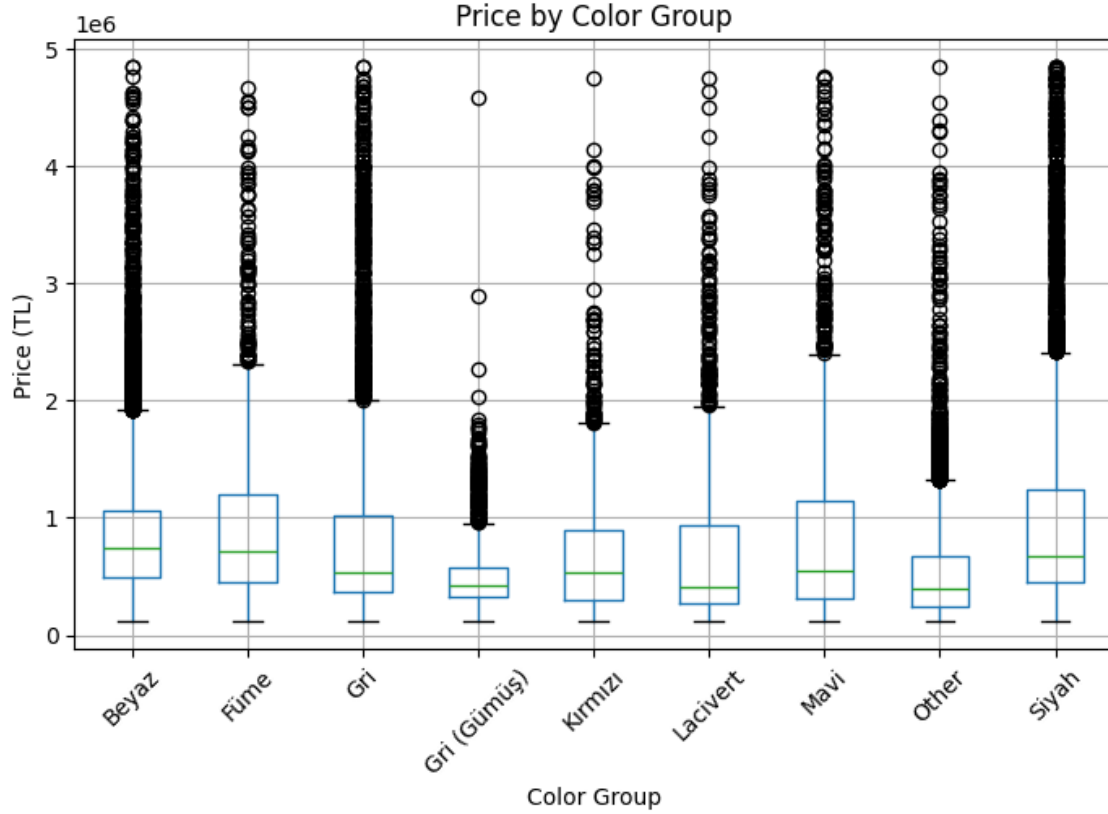


Figure 4. Distribution of estimated annual MTV values. Most listings fall into lower tax bands, while a smaller number of vehicles appear in much higher brackets.

The main EDA conclusions were consistent with expectations. First, price is highly skewed and benefits from a log transformation. Second, mileage is negatively associated with price, while newer vehicles tend to occupy higher price ranges. Third, structural variables such as transmission and body type produce clearer segmentation than color alone. Finally, although raw color differences do exist, those differences are not enough by themselves to establish a strong causal or practical color effect.

5. Hypothesis Testing

Two complementary hypothesis-testing strategies were used. The first asked whether price distributions differ across color groups at a purely descriptive level. The second asked whether color still adds explanatory power after structural variables are controlled for.

5.1 Kruskal–Wallis test on raw color groups

Because price distributions are not normally distributed, a Kruskal–Wallis test was used to compare color groups. The null hypothesis was that there is no significant difference in price distributions across color groups. The test returned a statistic of 4810.14 with $p < 0.001$, so the null hypothesis was rejected. This shows that raw price distributions differ across colors, but it does not account for structural vehicle characteristics.

5.2 Nested OLS comparison

The core hypothesis of the project was tested using two nested OLS regressions: a restricted model containing structural variables only, and a full model that added color group. Control variables included brand, series group, year, mileage, fuel type, transmission, and body type.

Statistic	Value
Kruskal–Wallis statistic	4810.14
Kruskal–Wallis p-value	< 0.001
Nested OLS p-value	2.14×10^{-14}
Restricted adjusted R^2	0.92648
Full adjusted R^2	0.92662
Δ adjusted R^2	0.00014

Table 2. Summary of the statistical hypothesis-testing results.

The nested comparison rejected the null hypothesis statistically: color does add explanatory power after structural variables are controlled for. However, the practical gain is tiny. The increase in adjusted R^2 is only 0.00014, which means that color contributes very little to the model’s overall explanatory power. In practical terms, the statistical tests support a nuanced conclusion: color is detectable, but weak.

6. Machine Learning Methods

The machine learning stage was designed to test the same substantive question from a predictive perspective: does adding color improve out-of-sample price prediction in a meaningful way? Two model families were trained: linear regression as a benchmark model and random forest as the main non-linear model. For each model family, two feature sets were compared: a baseline set containing structural variables only, and an extended set that additionally included color group.

The target variable was `log_price`. Model performance was evaluated on a held-out test set using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 . The feature sets were defined as follows. Baseline: year, log-transformed mileage, brand, series group, fuel type, transmission, body type group, and seller type. Extended: all baseline features plus color group.

Model	MAE	RMSE	R^2
Linear Regression - Baseline	110,409.67	228,834.96	0.8625
Linear Regression - Extended	110,484.17	228,551.40	0.8628
Random Forest - Baseline	122,067.09	257,389.30	0.8260
Random Forest - Extended	129,582.56	274,702.56	0.8018

Table 3. Machine learning model comparison on the held-out test set.

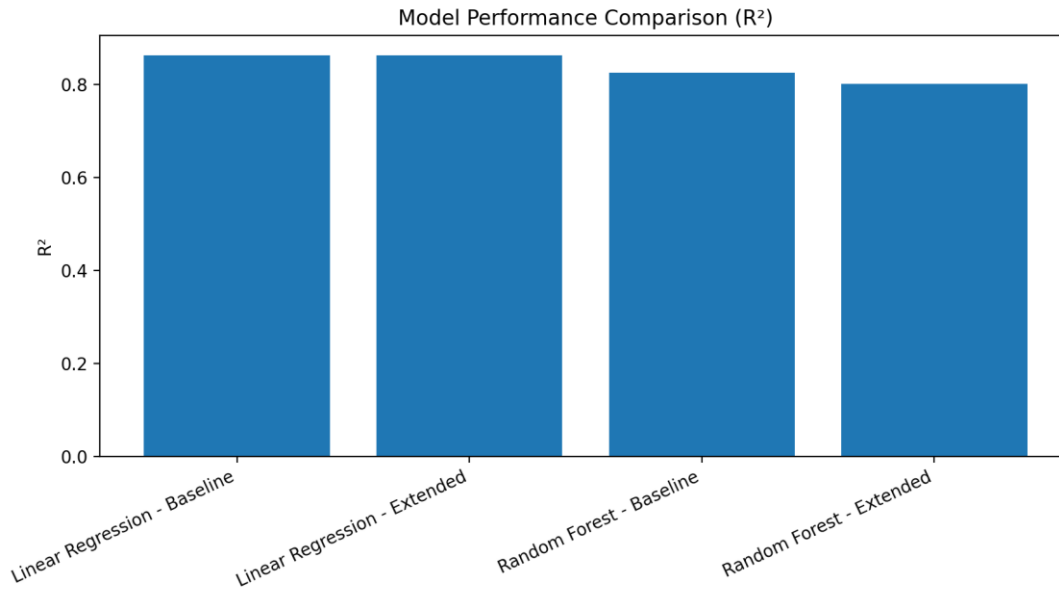


Figure 5. Model comparison using R^2 . Linear regression outperformed random forest on this dataset, and the gain from adding color was minimal.

The machine learning results reinforce the earlier statistical interpretation. The best overall performance came from the extended linear regression model, but its improvement over the baseline linear model was marginal. The increase in R^2 was only 0.00034, and the error metrics changed only slightly. This suggests that color may carry a small amount of predictive information, but its contribution is not substantively important.

Random forest performed worse than linear regression in this project. Moreover, adding color to random forest reduced performance rather than improving it. This does not mean that random forest is a poor model in general; instead, it suggests that under the current feature representation and parameterization, color does not provide stable predictive value. Taken together, the ML stage strengthens the conclusion that structural variables dominate price prediction.

7. Key Findings

- Used-car prices in the dataset are strongly right-skewed and are better modeled on the log scale.
- Mileage is negatively associated with price, while newer vehicles generally occupy higher price levels.
- Structural characteristics such as brand, series, year, mileage, transmission, fuel type, and body type explain most of the variation in prices.
- Raw color differences are statistically significant, but their practical effect is weak.
- Adding color to linear regression yields only a marginal predictive gain, while adding color to random forest worsens performance.
- The project hypothesis is supported overall: cosmetic variables such as color play only a limited marginal role relative to structural variables.

8. Limitations and Future Work

This project has several limitations. First, the dataset is cross-sectional and does not include listing dates, which prevents a robust integration of time-varying macroeconomic variables such as exchange rates and fuel prices. Second, the dataset does not include the official legal vehicle-value field used directly in the GİB tax schedule, so estimated MTV had to be constructed using listing price as a proxy. Third, the analysis is limited to the structure of one large online marketplace dataset and therefore may not capture the full diversity of the Turkish used-car market.

Future work could improve the project in several ways. Temporal data would allow price changes to be linked more convincingly to macroeconomic conditions. Additional market descriptors, more extensive hyperparameter tuning, and alternative algorithms such as gradient boosting could be tested. A richer causal design could also be developed by matching highly similar cars and varying only cosmetic attributes in a more controlled counterfactual setting.

9. Reproducibility and Repository

The project is organized in a public GitHub repository that contains the proposal PDF, a notebook for data collection/EDA/hypothesis testing, a separate notebook for machine learning, a requirements file, the dataset used in the notebooks, and an outputs directory for generated figures and tables. At the time of writing, the repository records 47 commits and includes a milestone2 release, which helps document the project's progression over the semester.

The repository structure supports transparent reproduction of the workflow. The README summarizes the project stages and the generated outputs folder stores the key visualizations and tabular results.

10. Conclusion

This project provides a data-driven analysis of used-car price formation in the Turkish second-hand market. Across exploratory analysis, formal statistical testing, and predictive modeling, the evidence points in the same direction: cosmetic attributes such as color can be statistically detected, but their practical importance is limited. The main determinants of price are structural characteristics — especially brand, model family, year, mileage, transmission, fuel type, and body type.

The final conclusion is therefore not that color is irrelevant, but that it is weak relative to the dominant structural variables. This distinction between statistical significance and practical significance is the central contribution of the project.

References

- Kaggle dataset based on arabam.com used-car listings.
- Revenue Administration of Türkiye (GİB), official 2026 Motor Vehicle Tax tariff and calculation resources.
- GitHub repository: derobtw/derinalc_dsa210.
- DSA 210 course proposal and project guideline documents.

AI Usage Disclosure

AI tools were used for assistance in code organization, debugging, explanatory writing, and report structuring. All final analytical decisions, implementation choices, and interpretations were made by the author.